

Jin Woo Oh

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SKILLS

Languages & Frameworks: Python, SQL, R, C++, React, TypeScript, JavaScript, dbt, Zero, Flask, Ollama, CUDA

Machine Learning & AI: Claude, Pandas, NumPy, Matplotlib, PyTorch, TensorFlow, Scikit-Learn (sklearn), HuggingFace, LangChain, Modal, fine-tuning, hypothesis testing, regression, anomaly detection, XGBoost, A/B testing, deep learning

Tools & DevOps: GitHub Actions, Docker, Kubernetes (k8s), AWS, GCP, Dagster, Temporal, Convex (NoSQL), PostgreSQL, Apache, Power BI, BigQuery, API, MCP, test automation (vitest, Playwright, pytest), Bash

EXPERIENCE

- Avnir** **Nov 2025 – Present**
Software Engineer *Atlanta, GA*
- Built Claude-powered contact-extraction pipeline using prompt engineering and structured JSON output, parallelizing large uploads by decomposing Temporal activity into concurrent batches, reducing ingestion time from 40 min to 4 min
 - Created an end-to-end relationship-strength scoring model (SOM) with 30 behavioral features tuned via sklearn grid search, stratified k-fold CV, and teacher-student score distillation; cut pipeline reruns from 75 minutes to 5 seconds via per-user caching
 - Optimized production GenAI services on k8s; migrated 3 vLLM servers (Qwen2.5→Gemma4) with GPU downsizing, eliminated cold-start failures via async warmup, and added KEDA autoscaling on queue backlog after root-causing 45-min pipeline jam
 - Automated Customer Success reporting pipeline by replacing a manual Google Sheet with 20-dataset/30-chart Apache Superset dashboard computing KPIs (DAU/WAU, engagement-depth, etc), deployed across 3 environments
 - Developed GeoPin, a location intelligence feature (self-hosted Nominatim on k8s) that recommends nearby contacts by goal-skill fit and finds low-resistance warm introduction path via Apache AGE graph on PostgreSQL; built the Python REST API backend (Clerk/OAuth) and React (TypeScript) map UI, exposed over MCP for in-house AI agents

- ICC Lab, Florida State University** **May 2025 – Nov 2025**
Researcher *Tallahassee, FL (Remote)*
- Automated 1,000+ hours of manual labeling by building a gaze-tracking pipeline with Tobii eye-tracking data and OpenCV to overlay real-time gaze positions and regions of interest on screen recordings to streamline behavioral analysis
 - Achieved 95% accuracy in detecting UI state change via robust template matching and HSV color masking

- FlightAware** **June 2024 – Aug 2024**
Software Engineer Intern *Houston, TX (Remote)*
- Reduced latency of deep learning web application by 35% by developing scalable API (k8s, Streamlit & FastAPI) to provide secure internal access to the web service across multiple teams and earning positive client feedback
 - Enhanced robustness and improved system performance of the Foresight Labs website by 20% via streamlining REST API calls tested on Postman, reconfiguring dependencies on Redis and Triton servers
 - Performed A/B test of GPT-4o vs Mistral 7B RAG chatbots; recommended GPT-4o for 21x speed & 40% higher success

- Samsung Electronics America** **Aug 2019 – Apr 2024**
Software Quality Engineer III *Berkeley Heights, NJ*
- Spearheaded the development of a GPU-accelerated technical classifier model to identify customer complaints citing critical issues of flagship models with over 92% accuracy, automating 11+ hours of work per week
 - Built and automated ETL pipeline (Python + SQL) processing 1M+ articles for downstream data classification and daily QA reporting of Android OS (9-14), improving data consistency and accelerating issue detection workflows by 25%
 - Reduced tech support live chat operation cost by 30% via a discrete-event simulation (Arena), optimizing staffing allocation
 - Presented clear, actionable business insights to stakeholders with user behavior & QA trend dashboards using Power BI

PROJECTS

[RayTracing](#) | C++, CUDA, linear algebra, computer graphics, parallel programming, GPU acceleration, multi-threading, HPC

- Implemented CUDA ray tracer achieving 35x GPU speedup over CPU via parallel programming and HPC scaling

[Technical Issue Classifier](#) | PyTorch, TensorFlow, Scikit-Learn, HuggingFace, CUDA, BeautifulSoup, Docker, k8s, SQL

- Optimized model by grid searching and threshold-tuning the ROC curve, minimizing costly false negatives to 3%

EDUCATION

- Georgia Institute of Technology** **Expected Dec 2026**
Master of Science (M.S.) in Analytics *Atlanta, GA (Online)*
- Seoul National University** **Feb 2018**
Master of Science (M.S.) in Mechanical Engineering *Seoul, Korea*
- 2 patents, 2 journal publications, 8 conference presentations; Academic Scholarship; 3.75/4.0 GPA, *Summa cum laude*
- University of Illinois at Urbana-Champaign** **May 2015**
Bachelor of Science (B.S.) in Mechanical Engineering *Champaign, IL*
- 1 journal publication; VP, Co-founder, mentor and tutor at Korean Mechanical Engineer Network; Academic Scholarship